

Computer Networks

Transmission Media (Unguided) and Antennas

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Antennas

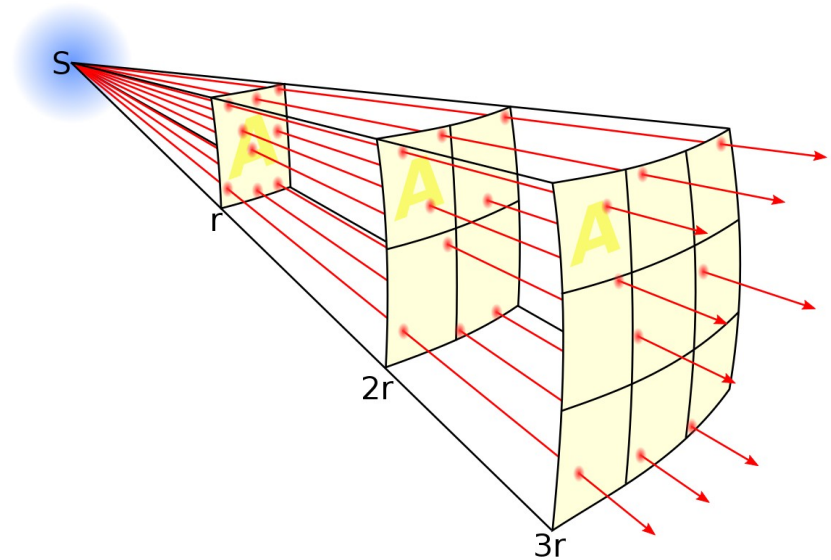
- ❑ Electrical conductors used to radiate or collect electromagnetic energy
 - ❑ **Transmission antenna:** Electrical energy → converted to electromagnetic energy → radiated into the surrounding
 - ❑ **Reception antenna:** Electromagnetic energy → converted to electrical energy → fed to the receiver
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Isotropic Antennas

□ Isotropic antenna:

- A point in space that radiates power in all directions equally with a spherical radiation pattern

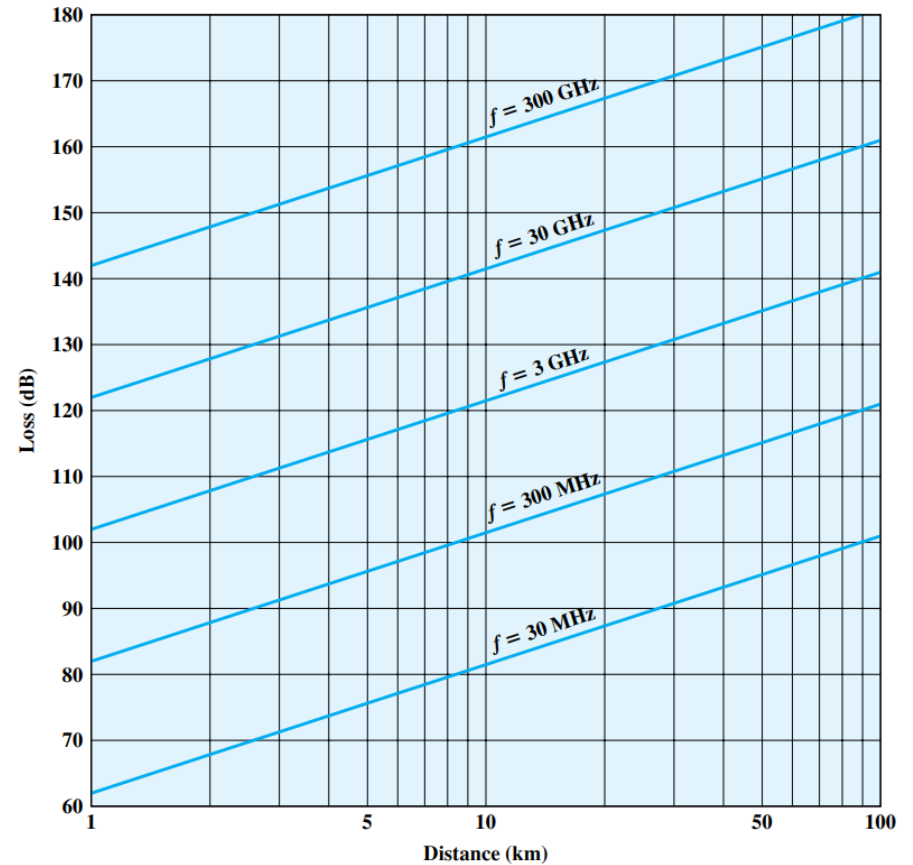
$$\frac{P_t}{P_r} = \left(\frac{4\pi d}{\lambda} \right)^2 = \left(\frac{4\pi f d}{c} \right)^2$$



Src: https://en.wikipedia.org/wiki/Free-space_path_loss

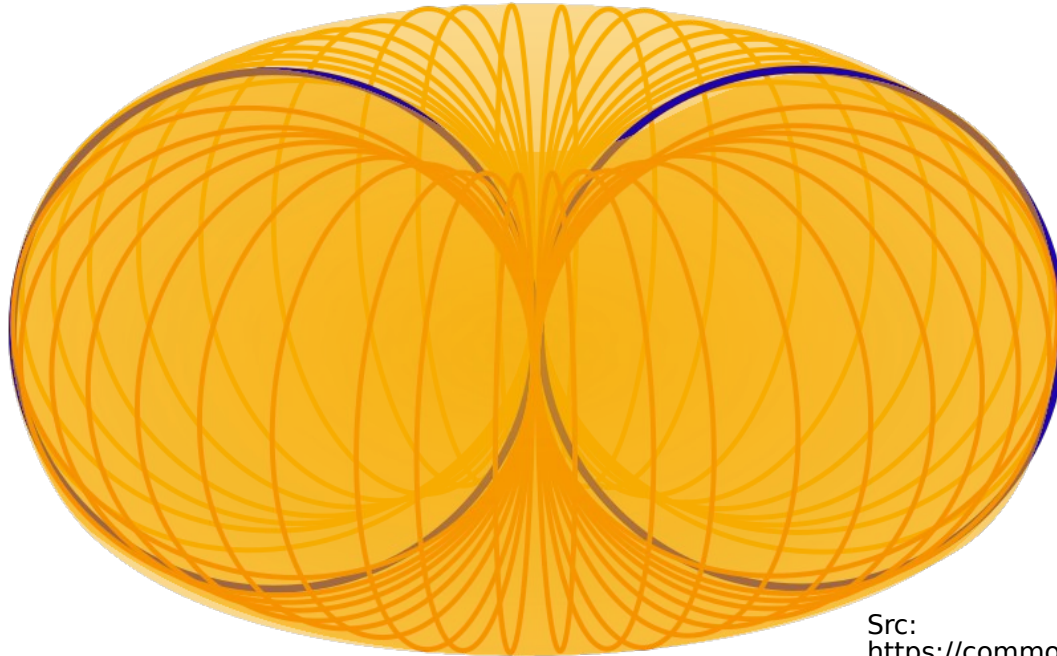
Isotropic Antennas

$$\frac{P_t}{P_r} = \left(\frac{4\pi d}{\lambda} \right)^2 = \left(\frac{4\pi f d}{c} \right)^2$$



Omni-directional Antennas

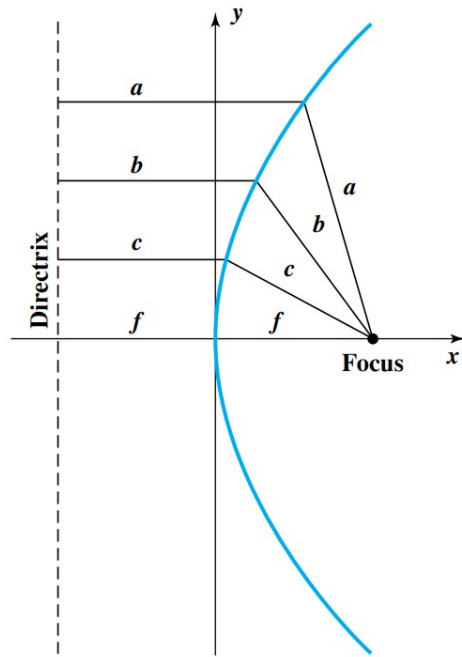
- ❑ Omni-directional antenna: power propagates uniformly in all directions in a plane
 - ❑ Cell phones, FM radios, walkie-talkies etc.



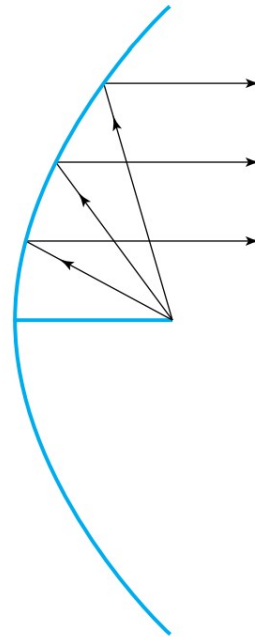
Src:
<https://commons.wikimedia.org/wiki/File:Element-doub-rad-pat-pers.svg>

Directional Antennas

□ Directional antenna: Parabolic reflective antenna



(a) Parabola



(b) Cross section of parabolic antenna showing reflective property

Antennas

- **Directional antenna:** Parabolic reflective antenna
 - Satellite communications, radio telescopes etc.

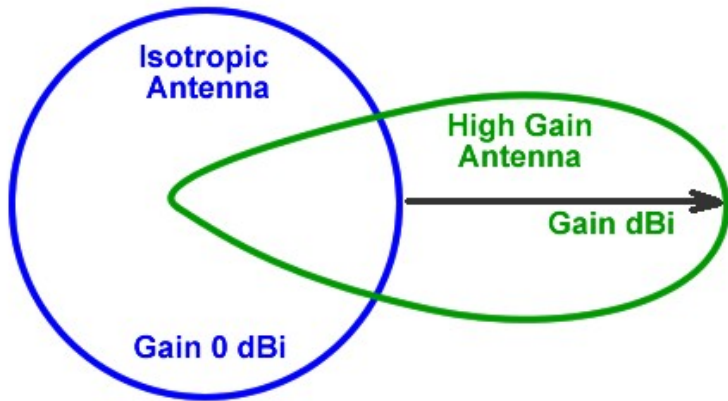


Src:https://commons.wikimedia.org/wiki/File:Antenna_03.JPG

Antenna Gain

□ Antenna gain:

- Measure of directionality
- Defined as the power output in a particular direction, compared to that produced in any direction by a perfect isotropic antenna (dBi)
- http://www.cisco.com/en/US/prod/collateral/wireless/ps7183/ps469/product_data_sheet0018630000078883b.html



Radiated power of isotropic antenna

$$G_{dB} = 10 \log_{10} \frac{P_i}{P_d}$$

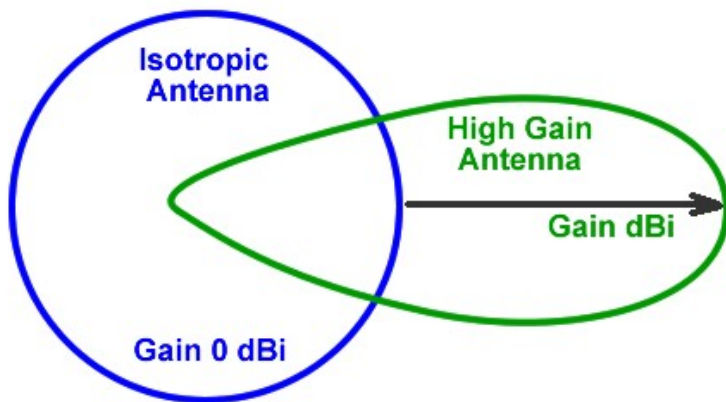
Radiated power of directional antenna

Src: <https://www.ahsystems.com/articles/Understanding-antenna-gain-beamwidth-directivity.php>

Antenna Gain

- Consider a directional antenna with a gain of 6 dB over a reference antenna and that radiates 700 W. How much power must the reference antenna radiates to provide the same signal power in the preferred direction?

$$G_{\text{dB}} = 10 \log_{10} \frac{P_i}{P_d}$$



Src: <https://www.ahsystems.com/articles/Understanding-antenna-gain-beamwidth-directivity.php>

Free Space Path Loss

Free space path loss:

$$\frac{P_t}{P_r} = \frac{1}{G_t G_r} \left(\frac{4\pi d}{\lambda} \right)^2 = \frac{1}{G_t G_r} \left(\frac{4\pi f d}{c} \right)^2$$

- G_t : Transmit antenna gain
 - G_r : Receiver antenna gain
 - Receiver antenna provides an aperture with an effective area for receiving a fraction of the transmitted power
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Free Space Path Loss

- Assume that a ground station is transmitting a signal of 250 W to a satellite at 4 GHz (earth to satellite distance is 35863 km). The antenna gains are 44 dB and 48 dB. What is the received power?

$$\frac{P_t}{P_r} = \frac{1}{G_t G_r} \left(\frac{4\pi d}{\lambda} \right)^2 = \frac{1}{G_t G_r} \left(\frac{4\pi f d}{c} \right)^2$$

Summary

□ Wireless transmission medium:

- Isotropic, omnidirectional and directional antenna
 - Free space path loss
 - Antenna gain
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